

1 • THE MODEL — LOG-LOGISTIC ACCELERATED FAILURE TIME

Duration is modelled on the log scale

$$\log T_i = \mu_i + \sigma W_i, \quad W_i \sim \text{Logistic}(0, 1), \quad \mu_i = o_i + \mathbf{x}_i^\top \boldsymbol{\beta}$$

 o_i = log planned duration (offset), so $e^{\hat{\theta}_0}$ reads directly as a multiplier of the plan.

Standardised residual

$$z_i = \frac{\log t_i - \mu_i}{\sigma}$$

Density and survival follow exactly

$$\begin{aligned} \log f(t_i) &= z_i - 2 \log(1 + e^{z_i}) - \log \sigma - \log t_i \\ \log S(t_i) &= -\log(1 + e^{z_i}) \end{aligned}$$

2 • FITTING — CENSORING IS EVIDENCE, NOT MISSING DATACensored log-likelihood ($\delta_i = 1$ if completed, 0 if still running)

$$\ell(\boldsymbol{\beta}, \sigma) = \sum_{i=1}^n [\delta_i \log f(t_i) + (1 - \delta_i) \log S(t_i)]$$

An unfinished project still says “not done by t_i ” — that is what $\log S$ encodes. Dropping it would bias the fit fast.

Penalised MLE — shrink thin sector/entity cells toward the pooled baseline

$$\hat{\theta} = \arg \min_{\theta} \left\{ -\ell(\theta) + \frac{1}{2\tau^2} \sum_{g \in G} \beta_g^2 \right\}$$

Equivalent to a $\mathcal{N}(0, \tau^2)$ prior on group effects. Solved by L-BFGS-B. τ by 5-fold CV on held-out, unpenalised log-likelihood

$$\tau^* = \arg \min_{\tau \in \mathcal{T}} \sum_{k=1}^5 \left[-\ell_{\text{unpen}}(\hat{\theta}^{(-k)}; \mathcal{D}_k) \right]$$

The penalty is a prior on parameters, so it must not enter the held-out score. An in-sample criterion picks the loosest prior every time.

3 • FROM FIT TO A CALIBRATED INTERVAL

Quantile function, then monotone rearrangement

$$Q(p) = m\left(\frac{p}{1-p}\right)^\gamma, \quad \tilde{Q}(p_j) = \max_{i \leq j} Q(p_i)$$

Split-conformal correction (Romano, Patterson & Candès 2019)

$$\begin{aligned} E_i &= \max(\hat{q}_{\text{lo}}(\mathbf{x}_i) - y_i, y_i - \hat{q}_{\text{hi}}(\mathbf{x}_i)) \\ Q &= E_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil, \quad C(\mathbf{x}) = [\hat{q}_{\text{lo}} - Q, \hat{q}_{\text{hi}} + Q] \end{aligned}$$

The $\lceil (n+1)(1-\alpha) \rceil$ rank is the finite-sample form; a plain empirical quantile under-covers.**MEASURED ON THE CORPUS**

Observations fitted	2,148
Observed completions $\sum \delta_i$	160
Right-censored	1,988
Event rate	7.4 %
Selected τ^* (5-fold CV)	0.20
Fitted scale $\hat{\sigma}$	0.332
Baseline multiplier $e^{\hat{\theta}_0}$	3.472 ×
Roads & Highways (145 events)	2.70 ×

CALIBRATION — MEASURED, NOT ASSERTED

Target coverage $1 - \alpha$	90.0 %
Empirical coverage (450 held-out)	93.3 %
Uncalibrated would be	84.7 %
Conformal quantile Q	5.185 mo
Split (train / cal / test)	900/450/450

WHAT THIS DOES NOT ESTABLISH

- **18 of 19 sectors are prior-dominated** — at a 7.4 % event rate they sit near the pooled baseline because the prior put them there. Only Roads moves on its own evidence. Each group publishes its event count and tag.
- The corpus records **no actual completion date** for an ongoing project. The 93.3 % belongs to the *slippage* interval; a coverage figure for the served P10–P95 date fan cannot be produced at all.
- Baseline of 3.47× exceeds the 1.64× median of finished projects. That is the censoring correction working — finishers are a biased-fast subsample.

Why it is defensible: the likelihood is the textbook log-logistic AFT with censoring; parameter recovery on synthetic data is unbiased to **0.022** even at 92.6 % censoring; and the interval carries a *measured* coverage number, not a promised one.

1 • MAKE THE NETWORK ACYCLIC BEFORE SCHEDULING IT

Tarjan strongly-connected-component condensation

$$G = (V, E) \xrightarrow{\text{SCC}} G^* = (V/\sim, E^*)$$

Mutually-dependent projects collapse to one node, so G^* is a DAG and a topological order exists. Verified acyclic at run time.

2 • SCHEDULE IN THE (MAX, +) SEMIRING

Forward pass — earliest start and finish

$$ES_j = \max_{i \rightarrow j} (EF_i + \ell_{ij}), \quad EF_j = ES_j + d_j$$

In max-plus algebra $\oplus = \max$, $\otimes = +$: the recursion is a linear system over that semiring.

Backward pass — per weakly-connected component, not one global finish

$$LF_j = \min_{j \rightarrow k} (LS_k - \ell_{jk}), \quad LS_j = LF_j - d_j$$

A forest of programmes floated against one global finish gave a median float of 206 years and made absorption inert.

Two different kinds of slack

$$TF_j = \max (LS_j - ES_j, 0) \quad FF_j = \max \left(\min_{j \rightarrow k} (ES_k - \ell_{jk}) - EF_j, 0 \right)$$

Total float = delay the whole chain can take. Free float = delay that reaches no successor.

Delay absorption and propagation

$$a_j = \min(\Delta_j, FF_j), \quad p_j = \max(\Delta_j - FF_j, 0)$$

Capital locked by a delay

$$L_j^{(50)} = c_j \cdot \frac{\Delta_j}{48}, \quad L_j^{(95)} = 1.65 L_j^{(50)}$$

3 • CRITICALITY — AN EXACT SHAPLEY VALUE ON THE REACHABILITY GAME

Reach set and coalition value

$$R(i) = \text{desc}(i) \cup \{i\}, \quad v(S) = \sum_{d \in \bigcup_{i \in S} R(i)} c_d$$

Shapley value, estimated by permutation Monte Carlo

$$\varphi_j = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \left[v(P_\pi^j \cup \{j\}) - v(P_\pi^j) \right]$$

 P_π^j = players before j in order π . $M = 30$ sampled orders, seeded.

Efficiency holds by construction — marginals telescope

$$\sum_{j \in V} \varphi_j = v(V)$$

MEASURED ON THE CORPUS

Nodes / edges	2,207 / 1,197
Acyclic DAG verified	yes
Isolated projects	926
$v(V)$ — full reachable capital	₹47.44 L Cr
Rescaled to locked capital	₹23.97 L Cr
Permutations M	30

CORRECTNESS CHECKS THAT WERE ACTUALLY RUN

- Exact agreement with enumerated Shapley on a 6-node DAG — max error $0.00e+00$ across all 720 permutations.
- Marginals telescope: per-permutation total is constant and equals $v(V)$ (spread 0.00).
- Isolated project $\Rightarrow \varphi$ = its own cost (it blocks only itself). Max error 0.005 Cr over 400 nodes.
- Rank-stable at $M = 30$: Spearman $\rho \approx 0.998$ across seeds, identical arg-max against $M = 2000$.

BOUNDS ON THE CLAIM

- Efficiency is a property of the *estimator*. The final rescaling into locked-rupee units is a **unit conversion, not evidence of the axiom** — dividing any positive numbers by their sum would satisfy it trivially.
- Edges are **declared dependency types** (statutory, raw-material, spatial-corridor, power-evacuation), not causality inferred from data.
- $M = 30$ is adequate for the *ordering* the product uses; it is not enough to quote an individual φ_j to two decimal places.

1 • DECISION VARIABLES

$$x_{i,t} \geq 0 \quad (i = 1 \dots N, t = 1, 2, 3), \quad \eta \in \mathbb{R}, \quad z_s \geq 0 \quad (s = 1 \dots S)$$

$x_{i,t}$ = capital to project i in tranche t . η, z_s are the Rockafellar–Uryasev auxiliaries. **Every variable is continuous — this is an LP, not a MILP.**

2 • OBJECTIVE — EXPECTED YIELD AGAINST TAIL RISK

$$\max_{x, \eta, z} (1 - \kappa) \sum_i \sum_t y_i m_t x_{i,t} - \kappa \text{CVaR}_\beta$$

y_i = completion yield per crore, m_t = tranche multiplier, $\kappa \in [0, 1]$ = the risk dial the judge can drag.

Rockafellar–Uryasev representation of Conditional Value-at-Risk

$$\text{CVaR}_\beta(L) = \min_{\eta} \left\{ \eta + \frac{1}{1 - \beta} \mathbb{E}[(L - \eta)^+] \right\}$$

The inner minimisation is why CVaR can enter an LP at all: the $(\cdot)^+$ linearises.

Linearised with scenario auxiliaries

$$\min \quad \eta + \frac{1}{1 - \beta} \sum_s p_s z_s \quad \text{s.t.} \quad z_s \geq L_s - \eta, \quad z_s \geq 0$$

3 • CONSTRAINTS

$$\begin{aligned} \sum_{i,t} x_{i,t} &\leq B && \text{(budget)} \\ \sum_{i \in \text{NER}} \sum_t x_{i,t} &\geq 0.10 B && \text{(statutory North-East floor)} \\ 0 \leq x_{i,t} &\leq \bar{u}_{i,t} && \text{(absorptive capacity)} \\ x_{i,t+1} \bar{u}_{i,t} &\leq x_{i,t} \bar{u}_{i,t+1} && \text{(SOS2 tranche ordering)} \end{aligned}$$

The last row enforces “fill tranche t before opening $t+1$ ” without a binary variable.

4 • SHADOW PRICES — WHAT THE DUALS MEAN

$$\pi_B = \frac{\partial \text{objective}}{\partial B}, \quad \pi_B \cdot \left(B - \sum_{i,t} x_{i,t} \right) = 0$$

Complementary slackness: $\pi_B > 0$ if and only if the budget is actually binding. Taken from a `linprog(method="highs")` solve — never synthesised.

MODEL CONSTANTS

CVaR confidence β	0.90
Tranches T	3
Scenarios S , yields θ_s	3 • 1.05/0.95/0.80
NER floor	10 % hard
Solver	HiGHS simplex
Solve time	< 10 ms

 π_B MEASURED AT $\kappa = 0.5$ — DIMINISHING RETURNS

Budget ₹2,000 Cr	2.233
Budget ₹10,000 Cr	1.216
Budget ₹20,000 Cr	0.773
Budget ₹34,000 Cr	0.432
Beyond $\approx$ ₹34,919 Cr	0.000

Past $\approx$ ₹34.9 k Cr the **budget stops binding — absorptive capacity does**. A zero shadow price there is the correct answer, and it is a finding: the portfolio cannot absorb more money that year.

STATED PRECISELY

- It is an **LP, not a MILP**. Capital tranches are genuinely divisible, so the continuous relaxation is the correct model — but the label had to match the mathematics.
- A shadow price is **meaningless without its budget level**. Quoting one number alone is how “ $\pi = 0.699$ ” survived in the docs after the value had moved.
- If the dual solve fails, **no shadow price is reported**. An invented dual is worse than a missing one — it is the number an officer would reallocate on.

1 • RESOLUTION IS COMPUTED, NEVER ASSERTED

Web Mercator ground sample distance

$$g(\phi, z) = \frac{156543.03392 \cos \phi}{2^z} \text{ m/px}$$

At $z = 16$ across India (8°N–34°N) this gives **1.98–2.37 m/px**. Every resolution claim in the product is this number, not a datasheet figure.

2 • MAKE TWO EPOCHS RADIOMETRICALLY COMPARABLE

Relative radiometric normalisation on pseudo-invariant features

$$A' = a A + b, \quad (a, b) = \arg \min_{a, b} \sum_{p \in \text{PIF}} (a A_p + b - B_p)^2$$

Fitted only on the 40 % most stable pixels, 4 iterations. Histogram matching was rejected: it forces the whole distribution to agree and so normalises the real change away.

3 • VISIBLE-BAND INDICES — THE BANDS THAT EXIST

Excess Green (Woebbecke et al. 1995)

$$\text{ExG} = \text{clip}(2g - r - b, -1, 1), \quad r, g, b \in [0, 1]$$

Visible Atmospherically Resistant Index (Gitelson et al. 2002)

$$\text{VARI} = \frac{g - r}{g + r - b}$$

Chosen because the haze term largely cancels between visible bands — the right index when no atmospheric correction is available.

Spectral Angle Mapper — illumination-invariant by construction

$$\theta = \arccos \left(\frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\| \|\mathbf{b}\|} \right)$$

A pixel that is merely *brighter* keeps its direction in RGB space ($\theta \approx 0$); a pixel whose material changed rotates. That is exactly the “sun angle moved” vs “the ground changed” discrimination.

4 • EPOCH SELECTION AND CHANGE

$$d_H(\text{phash}_i, \text{phash}_j) \geq 8 \implies \text{genuinely distinct captures}$$

$$\text{surface change} = \frac{|\{\text{structurally changed, vegetation-masked}\}|}{|\{\text{valid}\}|} \times 100$$

“Did anything actually happen on the ground during the period that was claimed?”

MEASURED

Wayback releases probed	196
Dated epochs resolved	26
Showcase sites	5
Analysis GSD (z16)	2.08–2.35 m/px
z18 display bake	0.539 m/px
Sites qualifying for z18	2 of 70
Per-image integrity	SHA-256

THE ONE STATISTIC THAT GOVERNS THIS ENGINE

$$r(\text{surface change, reported progress}) = 0.007$$

Statistically indistinguishable from noise. Imagery explains essentially **none** of the variance in reported progress — so a “satellite says 38.4 % complete” figure would be a guess wearing the costume of a measurement. **That figure existed here once and was removed:** it was 45 %-weighted on the very claim it purported to audit, which is why it looked plausible.

WHAT IT REPORTS INSTEAD — EVIDENCE, NEVER GUILT

- GROUND_ACTIVITY_OBSERVED
- NO_ACTIVITY_DETECTED — the only actionable verdict; it says “inspect”, never “fraud”
- SENSOR_CANNOT_RESOLVE — tunnels, interiors, underground
- NO_IMAGERY_FOR_INTERVAL
- NOT_MATERIAL — claimed delta too small to expect a signal

REFUSED IN CODE, NOT JUST IN THE PITCH

- Sub-metre change detection — measured GSD is 2.08–2.35 m/px.
- NDVI / NDBI / NDWI — the basemap is RGB only; no NIR or SWIR band exists to compute them.
- A “fine-tuned” Prithvi model — no labelled Indian construction-stage corpus exists; distant supervision is used and named as such.

1 • THE QUANTITY BEING EXAMINED

Cost overrun against the original Cabinet approval

$$o_i = \frac{C_i^{\text{rev}} - C_i^{\text{orig}}}{C_i^{\text{orig}}} \times 100$$

A revision crossing $o_i = 20\%$ triggers mandatory CCEA re-appraisal. That threshold is the discontinuity of interest.

2 • BOUNDARY BIN-MASS RATIO, WITH ITS UNCERTAINTY

Ratio of densities either side of the threshold

$$R = \frac{n_1}{n_2}, \quad n_1 = |\{o_i \in [18, 20)\}|, \quad n_2 = |\{o_i \in [20, 22)\}|$$

Log-normal standard error for a ratio of counts

$$SE_{\log} = \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

95 % confidence interval

$$CI_{95} = \left[e^{\ln R - 1.96 SE_{\log}}, e^{\ln R + 1.96 SE_{\log}} \right]$$

3 • STATUTORY ESCALATION CAP — CPWD GCC CLAUSE 10CC

Only 85 % of contract value is escalable; 15 % is retained contractor risk

$$\text{Escalable base} = 0.85 C^{\text{orig}}, \quad \text{Risk retention} = 0.15 C^{\text{orig}}$$

Component index growth since sanction year

$$\Delta I_c = \frac{I_{c,t}}{I_{c,0}} - 1$$

Statutorily allowable escalation, then the excess claim

$$E_{\text{allowed}} = 0.85 C^{\text{orig}} \sum_c w_c \Delta I_c$$

$$\text{Excess} = \max\left(0, C^{\text{rev}} - (C^{\text{orig}} + E_{\text{allowed}})\right)$$

Weights w_c : steel 0.20, cement 0.15, fuel 0.15, labour 0.25, other 0.25. Indices from the DPIIT/RBI WPI series.

“Do cost revisions cluster just under the limit that would trigger Cabinet review?”

MEASURED ON THE ACTIVE REVISED COHORT

Active revised population	1,183
n_1 — revisions in [18 %, 20 %)	28
n_2 — revisions in [20 %, 22 %)	17
Bin-mass ratio R	1.65 ×
$SE_{\log} = \sqrt{1/28 + 1/17}$	0.307
95 % CI	[0.90, 3.01]

READ THE INTERVAL BEFORE QUOTING THE RATIO

The 95 % interval **contains 1.0**. At $n_1 + n_2 = 45$, a ratio of 1.65× is **not statistically significant at the 5 % level**. We say so on the sheet rather than waiting to be asked.No p-value is computed anywhere in this system, and any claim of “ $p < 0.001$ ” for this test is withdrawn — it is arithmetically impossible alongside an interval that spans 1.0.The honest sentence: “1.65× more revisions land just below the Cabinet threshold than just above — 28 against 17, 95 % CI [0.90, 3.01]. Suggestive, not significant, and offered as an *audit-triage trigger*, not a finding.”

WHAT THIS IS NOT

- **Not an allegation.** Boundary proximity is a screening indicator. Documentary review is required to determine why any individual revision landed where it did.
- **Not a McCrary test in the formal sense.** This is a two-bin mass ratio with a log-normal interval — not a local-polynomial density-discontinuity estimator, and it is not labelled as one on this sheet.
- The Clause 10CC cap is a **statutory arithmetic check**, not a model. It has no error bar because it has no estimation step.

1 • THE MERKLE TREE — RFC 6962 EXACTLY

Empty tree, single leaf, and the recursion

$$\begin{aligned} \text{MTH}(\emptyset) &= \text{SHA-256}(\text{""}) \\ \text{MTH}(\{d_1\}) &= \text{SHA-256}(0x00 \parallel d_1) \\ \text{MTH}(D_n) &= \text{SHA-256}(0x01 \parallel \text{MTH}(D_{[0:k]}) \parallel \text{MTH}(D_{[k:n]})) \\ k &= \text{largest power of two} < n \end{aligned}$$

Why the two prefixes, and why that split

0x00 / 0x01 domain separation — leaves and internal nodes are drawn from disjoint hash domains, so an internal node can never be presented as a leaf.

Split at the largest power of two — avoids duplicating an odd final node, which is **CVE-2012-2459**: two different leaf sets yielding one root, at which point a matching root proves nothing.

2 • INCLUSION PROOF — THE AUDITOR'S CHECK

Fold the sibling path from the leaf upward

$$h_0 = \text{SHA-256}(0x00 \parallel d), \quad h_j = \begin{cases} \text{SHA-256}(0x01 \parallel s_j \parallel h_{j-1}) & s_j \text{ on the left} \\ \text{SHA-256}(0x01 \parallel h_{j-1} \parallel s_j) & s_j \text{ on the right} \end{cases}$$

accept $\iff$ **h_m = published root**

A malformed proof — non-hex sibling, wrong length, wrong type — is a **verification failure, never an exception**. This endpoint is pointed at briefings an auditor already suspects, so it must fail closed.

3 • SEALING THE CORPUS, AND CHAINING EVERY REVISION

Canonical row serialisation, then the content address

$$\text{root} = \text{MTH}([\text{canon}(\text{row}_1), \dots, \text{canon}(\text{row}_n)])$$

canon($\cdot$) follows **RFC 8785** JSON Canonicalisation, with dates normalised day-first — so a CSV bootstrap and a PostgreSQL read produce the *same* root.

Append-only revision chain

$$h_i = \text{SHA-256}(\text{canon}(\text{row}_i) \parallel h_{i-1}), \quad h_0 = 0$$

Enforced three ways: **REVOKE** on UPDATE/DELETE, a database trigger, and this hash chain. Breaking one is not enough.

THE SEALED CORPUS

Rows sealed	2,207
Columns canonicalised	25
Corpus version	1

Merkle root

```
e659bf58f2a62e7d7f530792b54b3159
ad68a464e7cbf956c2f02bffb61558e2
```

VERIFIED AGAINST THE OFFICIAL SPECIFICATION

- **Exact match to the RFC 6962 / Certificate Transparency reference vectors** for tree sizes $n = 0 \dots 8$.
- Every inclusion proof verifies at $n = 2,207$ — and at 1, 2, 3, 7, 8, 255, 256, 257, 1000.
- Tamper detection: wrong proof, forged leaf, corrupted sibling and wrong root all rejected.
- **CVE-2012-2459 collision avoided**: $\text{MTH}([a, b, c]) \neq \text{MTH}([a, b, c, c])$.
- Malformed proofs return `False` — no exception path.

PRECISION ABOUT THE CITATIONS

- **RFC 6962** is the Merkle construction. **RFC 8785** is the JSON canonicalisation scheme used to serialise a row. An earlier revision cited RFC 8785 for the tree itself — **that citation was wrong and is withdrawn**.
- The tree proves *integrity and inclusion* — that a figure came from the sealed corpus unaltered. It does **not** prove the underlying government data is itself correct.

Zero-hallucination pipeline: Fact $\rightarrow$ Assertion $\rightarrow$ Render. Figures are extracted, asserted by rule, and assembled deterministically. **No language model ever produces a number** — with the API key cleared the copilot answers purely from locally computed, Merkle-signed facts.